

Class 10 Congruence

Goals

1. Understand the definition of the sets $\mathbf{Z}_n$
2. Be able to carry out modular addition.
3. Use to test for congruence in proofs.

Start with handout about test for congruence.

Definition: The set $\mathbf{Z}_n$ (read $\mathbf{Z}$ mod n) consists of all possible remainders when dividing by n . So $\mathbf{Z}_n = \{0, 1, 2, \dots, n-1\}$.

We define the operation of addition mod n on the set $\mathbf{Z}_n$ as follows.
For any a and b in $\mathbf{Z}_n$, $a \oplus b = (a+b) \bmod n$.

Example:

Fill in the mod 5 addition table for $\mathbf{Z}_5$

$\oplus$	0	1	2	3	4
0					
1					
2					
3					
4					